THE FORMULA, PART BY PART

Conditional probability

one rule: keep the part of A that lies inside B, then rescale by the size of B

what is asked

Probability of A given B

The chance of A once B is known to have occurred. The bar reads "given": everything to its right is the restricted world we now live in.

numerator

The part of A inside B

Only the outcomes where both A and B happen count. The rest of A lies outside the condition and is dropped.

$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

denominator

Rescale to the world of B

Dividing by $P(B)$ makes the probabilities inside B add up to one again. It is a rescaling, not a new law of probability.